Status	Finished
Started	Saturday, 3 October 2026, 3:44 PM
Completed	Saturday, 3 October 2026, 3:55 PM
Duration	10 mins 59 secs
Grade	**9.00** out of 10.00 (**90**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Given an array of integers `nums` which is sorted in ascending order, and an integer `target`, write a function to search `target` in `nums`. If `target` exists, then return its index. Otherwise, return `-1`.

You must write an algorithm with `O(log n)` runtime complexity.

Example 1:

```
Input: nums = [-1,0,3,5,9,12], target = 9
Output: 4
Explanation: 9 exists in nums and its index is 4
```

Example 2:

```
Input: nums = [-1,0,3,5,9,12], target = 2
Output: -1
Explanation: 2 does not exist in nums so return -1
```

Constraints:

- `1 <= nums.length <= 10^4`
- `-10^4 < nums[i], target < 10^4`
- All the integers in `nums` are **unique**.
- `nums` is sorted in ascending order.

For example:

Test	Result
`print(search([-1,0,3,5,9,12],9))`	4

Answer: (penalty regime: 0 %)

Reset answer

```python
def search(nums, target):
    left = 0
    right = len(nums) - 1

    while left <= right:
        mid = (left + right) // 2

        if nums[mid] == target:
            return mid
        elif nums[mid] < target:
            left = mid + 1
        else:
            right = mid - 1

    return -1
```

	Test	Expected	Got	
□	`print(search([-1,0,3,5,9,12],9))`	`4`	`4`	□
□	`print(search([-1,0,3,5,9,12],2))`	`-1`	`-1`	□

Passed all tests! □

Correct

Marks for this submission: 1.00/1.00.

Question **2**

Correct

Mark 1.00 out of 1.00

Two string values S1, S2 are passed as the input. The program must print first N characters present in S1 which are also present in S2.

Input Format:

The first line contains S1.
The second line contains S2.
The third line contains N.

Output Format:

The first line contains the N characters present in S1 which are also present in S2.

Boundary Conditions:

2 <= N <= 10
2 <= Length of S1, S2 <= 1000

Example Input/Output 1:

Input:

abcbde
cdefghbb
3

Output:

bcd

Note:

b occurs twice in common but must be printed only once.

Answer: (penalty regime: 0 %)

```
s1 = input().strip()
s2 = input().strip()
n = int(input())

common = set(s2)
result = []

for ch in s1:
    if ch in common and ch not in result:
        result.append(ch)

    if len(result) == n:
        break
```

```
14
15 print("".join(result))
```

	Input	Expected	Got	
□	abcbde cdefghbb 3	bcd	bcd	□

Passed all tests! □

Correct

Marks for this submission: 1.00/1.00.

Question **3**

Correct

Mark 1.00 out of 1.00

Given an list, find peak element in it. A peak element is an element that is greater than its neighbors.

An element a[i] is a peak element if

A[i-1] <= A[i] >=a[i+1] for middle elements. [0<i<n-1]

A[i-1] <= A[i] for last element [i=n-1]

A[i]>=A[i+1] for first element [i=0]

Input Format

The first line contains a single integer n , the length of A .
The second line contains n space-separated integers,A[i].

Output Format

Print peak numbers separated by space.

Sample Input

5

8 9 10 2 6

Sample Output

```
10 6
```

For example:

Input	Result
4 12 3 6 8	12 8

Answer: (penalty regime: 0 %)

```
n = int(input())
arr = list(map(int, input().split()))

peaks = []

for i in range(n):
    if i == 0:
        if arr[i] >= arr[i + 1]:
            peaks.append(arr[i])
    elif i == n - 1:
        if arr[i] >= arr[i - 1]:
            peaks.append(arr[i])
    else:
        if arr[i - 1] <= arr[i] >= arr[i + 1]:
            peaks.append(arr[i])

print(*peaks)
```

	Input	Expected	Got	
□	7 15 7 10 8 9 4 6	15 10 9 6	15 10 9 6	□
□	4 12 3 6 8	12 8	12 8	□

Passed all tests! □

Correct

Marks for this submission: 1.00/1.00.

Question **4**

Correct

Mark 1.00 out of 1.00

Given two Strings s1 and s2, remove all the characters from s1 which is present in s2.

Constraints

1<= string length <= 200

Sample Input 1

```
experience
enc
```

Sample Output 1

```
xpri
```

Answer: (penalty regime: 0 %)

```
s1 = input().strip()
s2 = input().strip()

remove = set(s2)

result = ""

for ch in s1:
    if ch not in remove:
        result += ch

print(result)
```

	Input	Expected	Got	
□	experience enc	xpri	xpri	□

Passed all tests! □

Correct

Marks for this submission: 1.00/1.00.

Question **5**

Incorrect

Mark 0.00 out of 1.00

String should contain only the words are not palindrome.

Sample Input 1

Malayalam is my mother tongue

Sample Output 1

is my mother tongue

Answer: (penalty regime: 0 %)

```
s = input().strip()

words = s.split()
result = []

for word in words:
    if word.lower() != word.lower()[::-1]:
        result.append(word)

print(" ".join(result))
```

	Input	Expected	Got	
☐	Malayalam is my mother tongue	is my mother tongue	is my mother tongue	☐

Your code failed one or more hidden tests.
Your code must pass all tests to earn any marks. Try again.

Incorrect

Marks for this submission: 0.00/1.00.

Question **6**

Correct

Mark 1.00 out of 1.00

Given an array `nums` containing `n` distinct numbers in the range `[0, n]`, return *the only number in the range that is missing from the array.*

Example 1:

```
Input: nums = [3,0,1]
Output: 2
Explanation: n = 3 since there are 3 numbers, so all numbers are in the range [0,3]. 2 is the missing number in the range since it does not appear in nums.
```

Example 2:

```
Input: nums = [0,1]
Output: 2
Explanation: n = 2 since there are 2 numbers, so all numbers are in the range [0,2]. 2 is the missing number in the range since it does not appear in nums.
```

Example 3:

```
Input: nums = [9,6,4,2,3,5,7,0,1]
Output: 8
Explanation: n = 9 since there are 9 numbers, so all numbers are in the range [0,9]. 8 is the missing number in the range since it does not appear in nums.
```

For example:

Test	Result
print(missingNumber([3,0,1]))	2
print(missingNumber([0,1]))	2

Answer: (penalty regime: 0 %)

Reset answer

```
def missingNumber(numbers):
    n = len(numbers)
    return n * (n + 1) // 2 - sum(numbers)
```

	Test	Expected	Got	
☐	`print(missingNumber([3,0,1]))`	2	2	☐
☐	`print(missingNumber([0,1]))`	2	2	☐
☐	`print(missingNumber([9,6,4,2,3,5,7,0,1]))`	8	8	☐

Passed all tests! ☐

Correct

Marks for this submission: 1.00/1.00.

Question **7**

Correct

Mark 1.00 out of 1.00

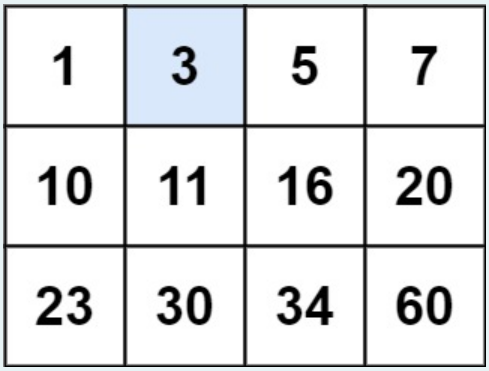
You are given an `m x n` integer matrix `matrix` with the following two properties:

- Each row is sorted in non-decreasing order.
- The first integer of each row is greater than the last integer of the previous row.

Given an integer `target`, return `True` *if* `target` *is in* `matrix` *or* `False` *otherwise*.

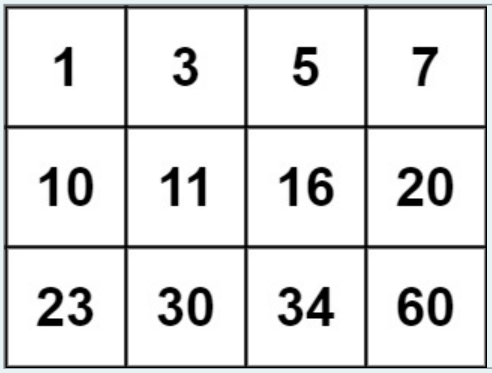
You must write a solution in `O(log(m * n))` time complexity.

Example 1:




```
Input: matrix = [[1,3,5,7],[10,11,16,20],[23,30,34,60]], target = 3
Output: True
```

Example 2:




```
Input: matrix = [[1,3,5,7],[10,11,16,20],[23,30,34,60]], target = 13
Output: False
```

For example:

Test	Result
print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 13))	False
print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 3))	True

Answer: (penalty regime: 0 %)

Reset answer

```
def searchMatrix(matrix, target):
    rows = len(matrix)
    cols = len(matrix[0])

    left = 0
```

```
    right = rows * cols - 1

    while left <= right:
        mid = (left + right) // 2

        value = matrix[mid // cols][mid % cols]

        if value == target:
            return True
        elif value < target:
            left = mid + 1
        else:
            right = mid - 1

    return False
```

	Test	Expected	Got	
□	print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 13))	False	False	□
□	print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 3))	True	True	□

Passed all tests! □

Correct

Marks for this submission: 1.00/1.00.

Question **8**

Correct

Mark 1.00 out of 1.00

Write a Python program for binary search.

For example:

Input	Result
1,2,3,5,8 6	False
3,5,9,45,42 42	True

Answer: (penalty regime: 0 %)

```
arr = list(map(int, input().split(',')))
target = int(input())

arr.sort()

left = 0
right = len(arr) - 1
found = False

while left <= right:
    mid = (left + right) // 2

    if arr[mid] == target:
        found = True
        break
    elif arr[mid] < target:
        left = mid + 1
    else:
        right = mid - 1

print("True" if found else "False")
```

	Input	Expected	Got	
☐	1,2,3,5,8 6	False	False	☐
☐	3,5,9,45,42 42	True	True	☐
☐	52,45,89,43,11 11	True	True	☐

Passed all tests! ☐

Correct

Marks for this submission: 1.00/1.00.

Question **9**

Correct

Mark 1.00 out of 1.00

Balanced strings are those that have an equal quantity of 'L' and 'R' characters.

Given a balanced string s, split it in the maximum amount of balanced strings.

Return the maximum amount of split balanced strings.

Example 1:

Input:

RLRRLLRLRL

Output:

4

Explanation: s can be split into "RL", "RRLL", "RL", "RL", each substring contains same number of 'L' and 'R'.

Example 2:

Input:

RLLLLRRRLR

Output:

3

Explanation: s can be split into "RL", "LLLRRR", "LR", each substring contains same number of 'L' and 'R'.

Example 3:

Input:

LLLLRRRR

Output:

1

Explanation: s can be split into "LLLLRRRR".

Constraints:

1 <= s.length <= 1000

s[i] is either 'L' or 'R'.

s is a balanced string.

For example:

Test	Result
`print(BalancedStrings('RLRRLLRLRL'))`	4
`print(BalancedStrings('RLLLLRRRLR'))`	3

Answer: (penalty regime: 0 %)

Reset answer

```
def BalancedStrings(s):
    balance = 0
    count = 0

    for ch in s:
        if ch == 'L':
            balance += 1
        else:
            balance -= 1

```

```
        if balance == 0:
            count += 1

    return count
```

	Test	Expected	Got	
☐	print(BalancedStrings('RLRRLLRLRL'))	4	4	☐
☐	print(BalancedStrings('RLLLLRRRLR'))	3	3	☐

Passed all tests! ☐

Correct

Marks for this submission: 1.00/1.00.

Question **10**

Correct

Mark 1.00 out of 1.00

An list contains N numbers and you want to determine whether two of the numbers sum to a given number K. For example, if the input is 8, 4, 1, 6 and K is 10, the answer is yes (4 and 6). A number may be used twice.

Input Format

The first line contains a single integer n , the length of list

The second line contains n space-separated integers, list[i].

The third line contains integer k.

Output Format

Print Yes or No.

Sample Input

7

0 1 2 4 6 5 3

1

Sample Output

Yes

For example:

Input	Result
5 8 9 12 15 3 11	Yes
6 2 9 21 32 43 43 1 4	No

Answer: (penalty regime: 0 %)

```
n = int(input())
arr = list(map(int, input().split()))
k = int(input())

seen = set()
found = False

for x in arr:
    if k - x in seen:
        found = True
        break

    if x * 2 == k:
        if x in seen:
            found = True
            break

    seen.add(x)

print("Yes" if found else "No")
```

	Input	Expected	Got	
☐	5 8 9 12 15 3 11	Yes	Yes	☐
☐	6 2 9 21 32 43 43 1 4	No	No	☐
☐	6 13 42 31 4 8 9 17	Yes	Yes	☐

Passed all tests! ☐

Correct

Marks for this submission: 1.00/1.00.